ACE 2026 | DGX Spark Mini Hackathon

PARTICIPANT CONNECTION GUIDE

NVIDIA Sync + GIGABYTE AI TOP ATOM

YOUR LAPTOP runs NVIDIA Sync and VS Code. YOUR ASSIGNED ATOM runs the remote coding workspace and the preconfigured local AI inference service. Follow the steps below; do not connect to other teams' ATOMs.

1. CONNECT TO THE WORKSHOP Wi-Fi

Connect to these WiFi to be on the same LAN as the GIGABYTE ATOM.

Wi-Fi: ACE 2026 - A / B 5G

Password: acedgx2026

Connect to the event network assigned to the Atoms by the organisers, not the separate venue guest network.
If you are not connecting to the Atom, you can connect to the public WiFi:

WiFi: PublicWifi@ASB
Password: ASB-CONNECTED

https://emrq.one/ace

2. INSTALL AND OPEN NVIDIA SYNC

Download NVIDIA Sync on your laptop:

https://build.nvidia.com/spark/connect-to-your-spark/sync

• Windows: Install the downloaded .exe. macOS: Open the .dmg and move NVIDIA Sync to Applications. Ubuntu: Follow NVIDIA's installation instructions at the link above.

• Launch NVIDIA Sync. On first use, agree to the EULA and choose your available development apps (for example VS Code or Cursor).

• Install VS Code or Cursor on your laptop beforehand if you wish to use it for remote coding. Terminal access is available without either IDE.

3. ADD YOUR ASSIGNED GIGABYTE ATOM

Open NVIDIA Sync → Settings (gear) → Devices → Add Device.

• Select your assigned ATOM if NVIDIA Sync discovers it, or choose “Add a device manually”.

• Enter your TEAM'S assigned device details:

Device name: Your team's GIGABYTE ATOM

Hostname pattern: EMARQUE-DGX-(ABCDEF).local

Example hostname: EMARQUE-DGX-C.local

SSH port: 22

Username: admin

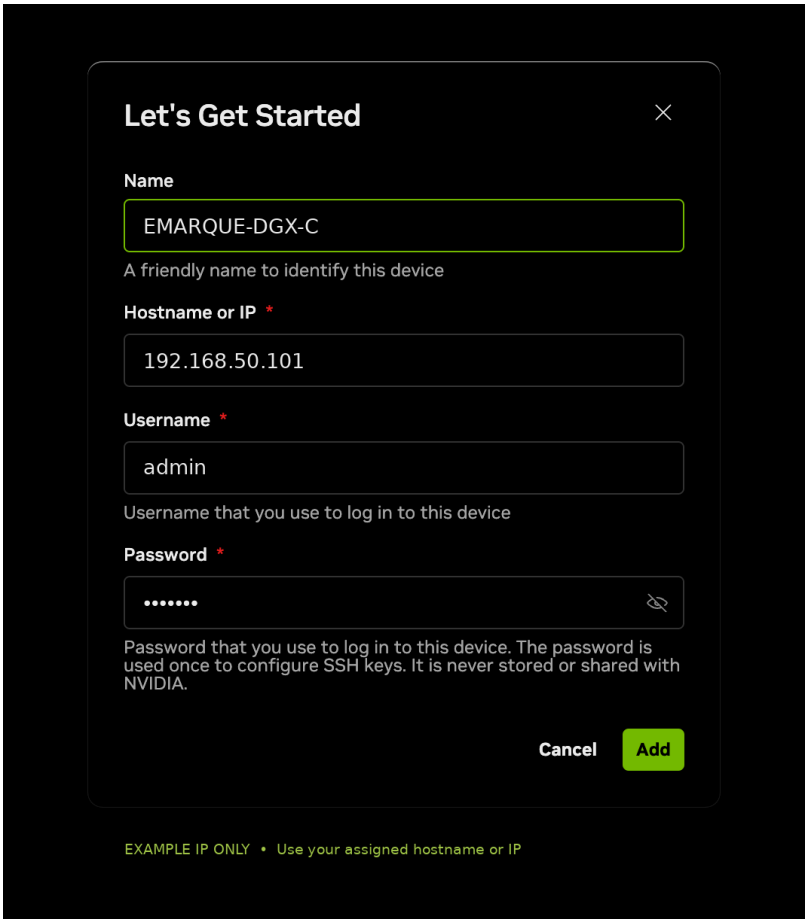
Password: Emarque123

• Click Add. If asked for permission to change Docker-group access or system configuration, consult a mentor before proceeding.

IMPORTANT: Please do not connect to other Atoms.

The hostname shown above is an example; follow your team assignment from the organisers. If .local discovery fails, use the assigned IP address provided by technical support.

NVIDIA Sync — Add your assigned ATOM (example)



The IP shown is an example only (192.168.50.101). Use your own assigned hostname or IP.

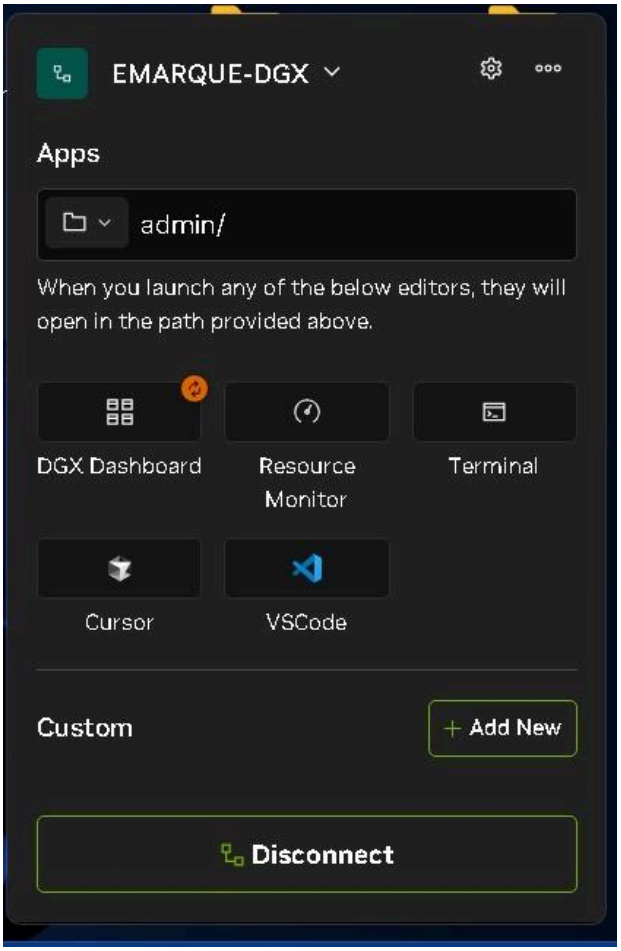
4. CONNECT AND LAUNCH YOUR DEVELOPMENT ENVIRONMENT

• In NVIDIA Sync, select your saved ATOM from the device dropdown and click Connect.

• Once connected, select Terminal, VS Code or Cursor (depending on what is installed).

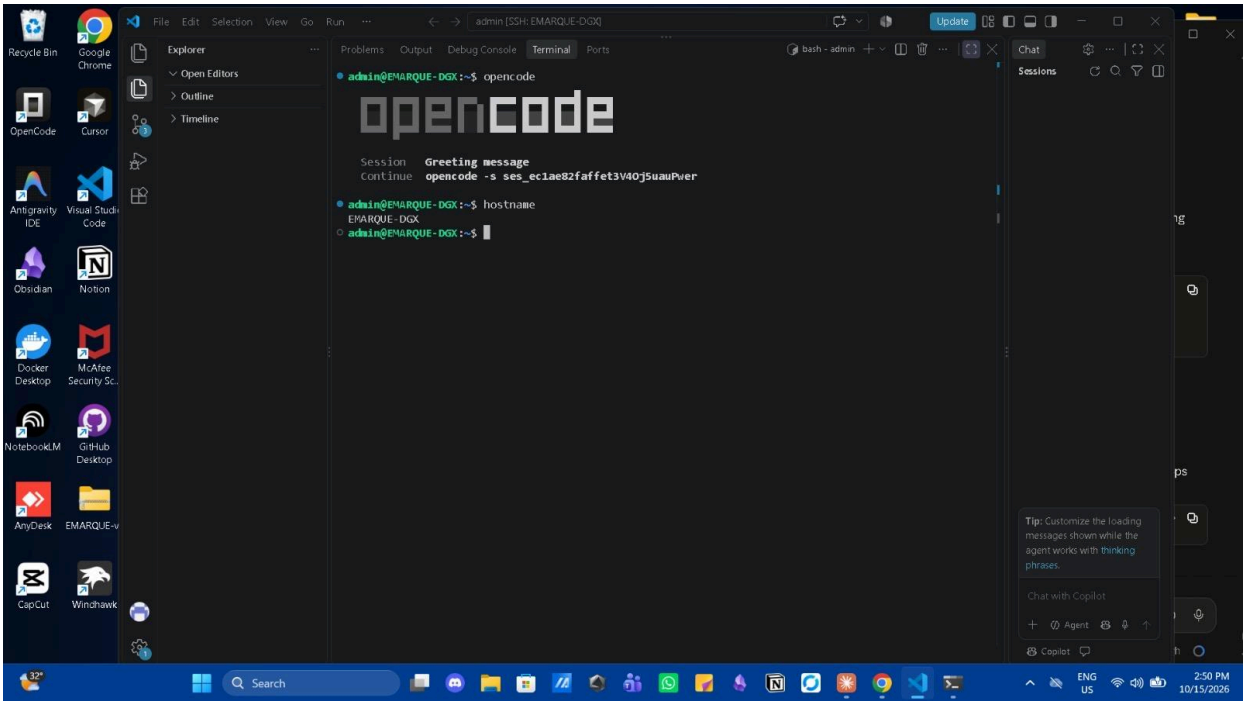
• Terminal opens an SSH session on the assigned ATOM. Remote IDEs let you work on files through the connection.

After connection — NVIDIA Sync management and applications



Shown: DGX Dashboard, Resource Monitor, Terminal, Cursor and VS Code on a connected ATOM.

REMOTE CODING — VS CODE VIA NVIDIA SYNC



Example: VS Code connected to EMARQUE-DGX over SSH, with OpenCode launched from its integrated terminal.

5. VERIFY YOUR ATOM AND LOCAL AI

In the NVIDIA Sync remote Terminal:

• Run: hostname

Verify that the hostname belongs to your assigned team.

• Run: nvidia-smi

Confirm the NVIDIA GPU environment is accessible.

• Confirm the local model endpoint is READY using the event-specific test command supplied by EMARQUE. The baseline model in the workshop briefing is NVIDIA Nemotron 3.5 Lightning; the actual served model ID must be verified by organisers.

• Do not download models, reinstall software or launch a second large inference service without approval.

6. START CODING WITH OPENCODE (DEFAULT)

OpenCode is a coding agent. It runs inside the remote ATOM terminal and can use the preconfigured local model served by vLLM. NVIDIA Sync itself does not provide the model.

• From NVIDIA Sync, launch VS Code or Terminal on your assigned ATOM.

• Open a terminal on the ATOM and navigate to your team's assigned project directory (provided by EMARQUE).

• Check installation: `opencode --version`

• Launch the agent: `opencode`

• Select the workshop's preconfigured LOCAL model/provider. If it is missing, ask a mentor; do not choose a cloud model to meet the local-inference requirement.

• First test prompt: “Create a minimal web app with an input field and results panel. Integrate it with our provided local inference endpoint. Explain how to run and test it.”

• Review proposed file changes and tool actions before approval. Use only synthetic or approved workshop data.

If opencode is not found, contact EMARQUE support. Do not spend workshop time installing the coding runtime or modifying the model server.

7. BUILD AS A FIVE-PERSON TEAM

• Nominate 1–2 members for NVIDIA Sync / ATOM access (coding agent and application integration).

• Other members work in parallel on frontend, tools/data, testing, documentation and integration using their laptops and team repository.

• Use the inference service already configured on your team's ATOM. NVIDIA Sync is a connection tool; it does not launch or load the model. Do not start another vLLM process.

• Agree on an integration owner so the prototype is tested and ready for submission within the three-hour hacking window.

OPTIONAL CODING TOOLS (IF PRECONFIGURED)

Cline (VS Code extension): An alternative for developers who prefer working inside the IDE. The organiser must provide the tested OpenAI-compatible model endpoint, model ID and any API key. Do not assume it is already configured.

Claude Code: Optional only when a participant has authorised access, or EMARQUE has tested an Anthropic-compatible local gateway. A normal Claude Code subscription or cloud API call does NOT, by itself, demonstrate local inference on the ATOM.

Cursor: Supported as a remote IDE through NVIDIA Sync; its built-in cloud AI is separate from the local vLLM model. The final project must still demonstrate inference on the assigned ATOM.

Keep the default path simple: NVIDIA Sync → VS Code/Terminal → OpenCode → preloaded local model. NeMo Switchyard and multi-model experiments are optional, not prerequisites.

TROUBLESHOOTING

ATOM does not appear

• Check that your laptop is on ACE 2026 - A / B 5G. Add the ATOM manually using its assigned IP address; some networks block mDNS / .local discovery.

Connection times out or is refused

• Confirm the assigned ATOM is online. Ask technical support to verify LAN access and SSH TCP port 22.

Authentication failed

• Recheck the assigned hostname, username and password. Ask tech support; do not reset system passwords.

VS Code or Cursor does not appear

• Check that the app is installed locally. Use NVIDIA Sync Terminal as a fallback.

General issues

• Raise your hand for the technical crew. Avoid changing the network, SSH, OS or inference-server configuration.

WORKSHOP RULES

• Access only your assigned GIGABYTE AI TOP ATOM; do not browse other teams' devices or files.

• Do not factory-reset, re-flash, rename, change passwords, or stop shared AI services.

• Keep work in your team's designated project directory and save the final code/workflow to your team's repository.

• Do not share workshop login credentials outside the event.

OFFICIAL NVIDIA REFERENCES

Install and connect: https://build.nvidia.com/spark/connect-to-your-spark/sync

Direct connections: https://docs.nvidia.com/sync/latest/direct-connections.html

Apps and remote development: https://docs.nvidia.com/sync/latest/applications.html